

TECHNOLOGY STACK

Date	26 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack:

This technology stack enables the system to provide real-time flood predictions, interactive map visualization, reliable data storage, and timely emergency alerts while remaining easy to maintain and expand in the future.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Jinja2 Templates	Provides a responsive and user-friendly web interface for entering weather parameters and displaying flood prediction results. Jinja2 enables dynamic rendering of prediction outputs.
2	Backend / Web Application	Python, Flask	Handles HTTP requests, manages application routing, validates user inputs, integrates the Machine Learning model, and returns prediction results to the frontend.

3	Machine Learning Layer	XGBoost, Scikit-learn	Performs flood prediction using the trained XGBoost classification model. Scikit-learn is used for preprocessing and feature scaling.
4	Data Preprocessing Layer	Pandas, NumPy, StandardScaler	Cleans, preprocesses, and transforms weather parameters before passing them to the Machine Learning model for prediction.
5	Model Storage Layer	Joblib	Stores the trained XGBoost model (flood_model.joblib) and StandardScaler (scaler.joblib) for efficient loading during runtime.
6	Development Environment	Jupyter Notebook, VS Code	Jupyter Notebook is used for model development and experimentation, while VS Code is used for Flask application development and debugging.
7	Deployment Layer	Render	Hosts the Flask web application on the cloud, making the Flood Prediction System accessible through a public URL.
8	Version Control	Git, GitHub	Manages source code, tracks project changes, and enables collaboration and cloud-based code hosting.